

CROWD MODERATION

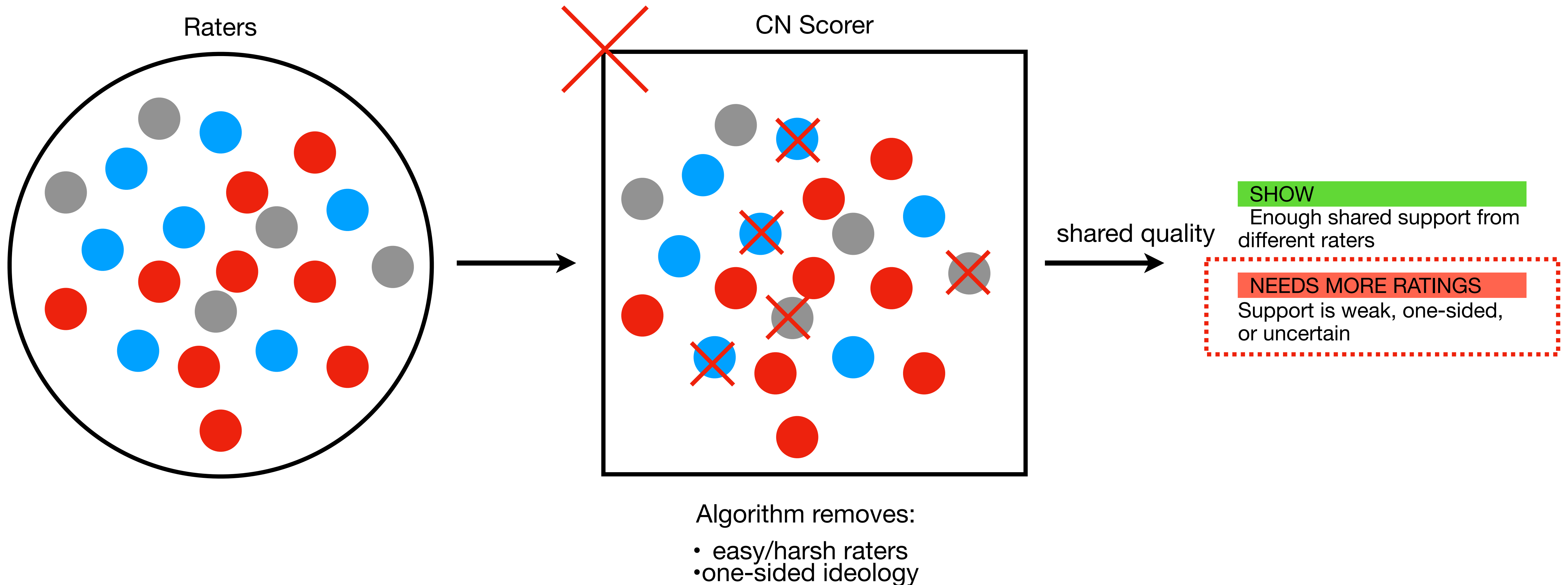
Cross-Constituency Aggregation (CCA)

How the display decision should aggregate ratings when disagreement is **structured and persistent** — in X's Community Notes.

Original algorithm

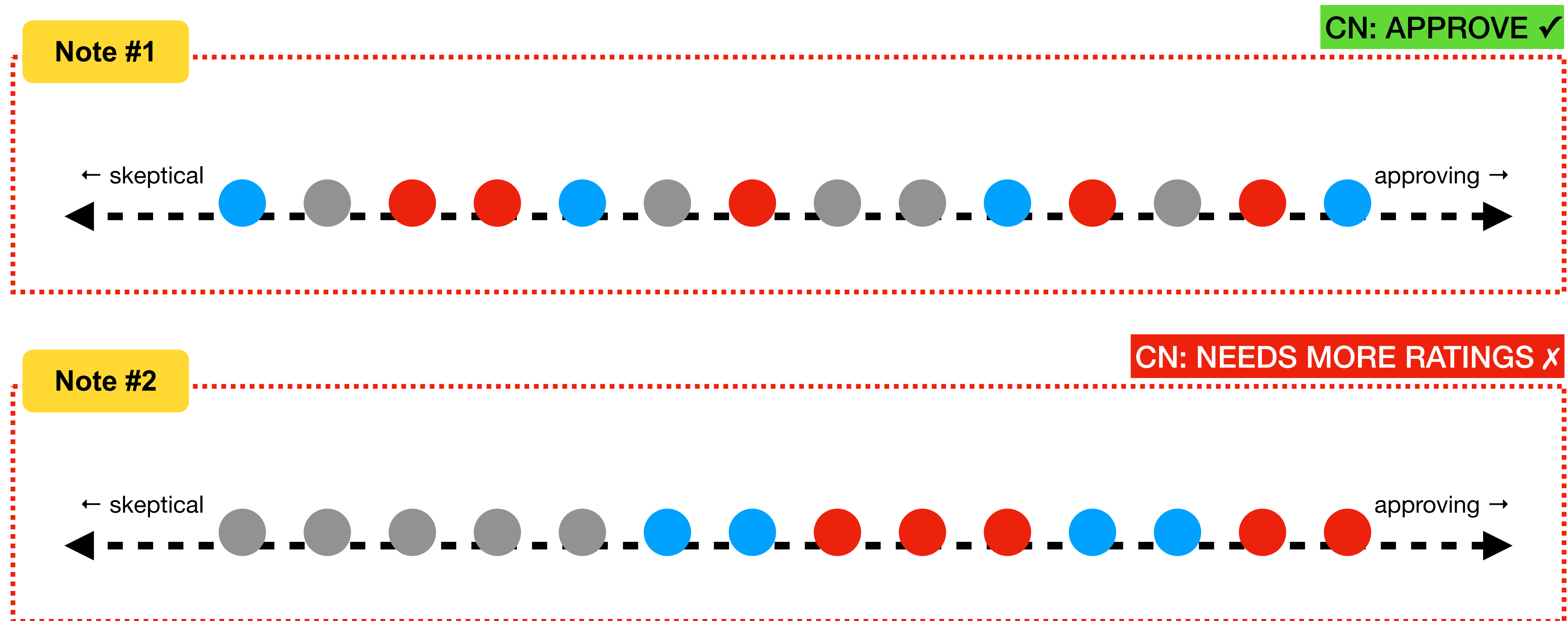
A note is shown only when support looks shared, not one-sided.

> CN tries to separate real note quality from rater bias and ideology.



THE GAP

CN looks for bridging, but still through one fitted axis



13,655 notes are stranded like Note #2.
Both groups wanted it shown. CN's axis couldn't see that.

OUR Goal

CN Algorithm

“What is the note’s real quality
after removing ideology?”

CN looks for one
hidden quality score.

CCA Algorithm

“Does each group accept the note
on its own terms?”

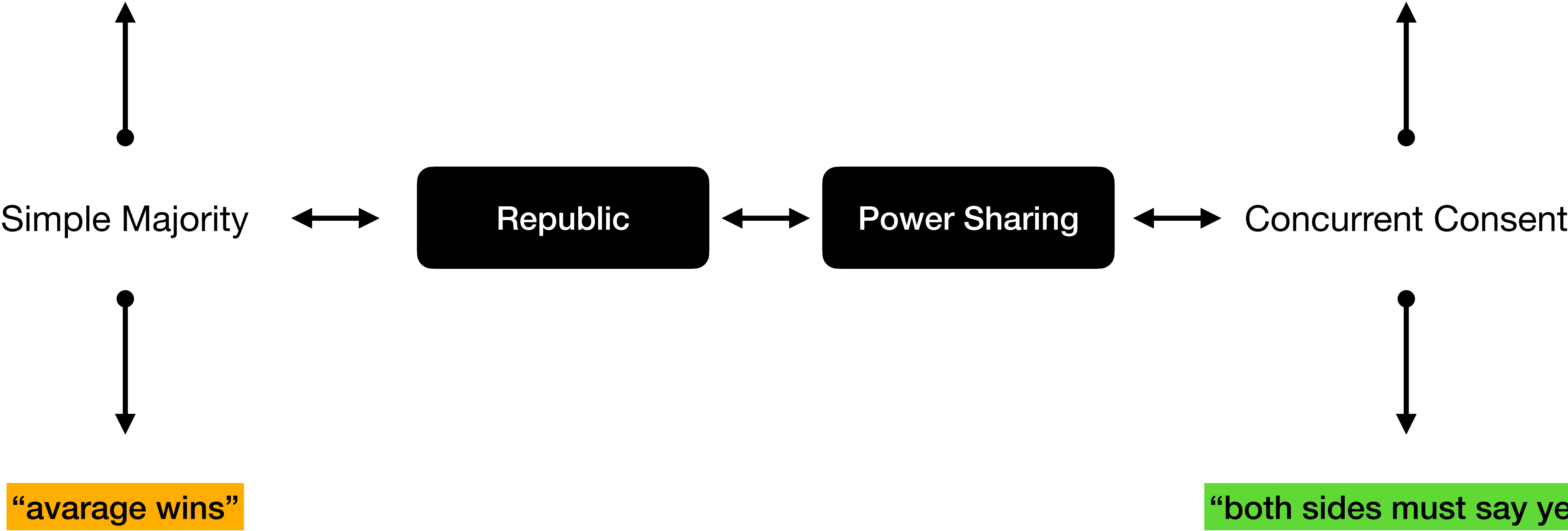
We measure approval within
each ideology.

RQ: What aggregation rule is appropriate when
disagreement is structured and persistent?

Inspiration

shallow disagreement

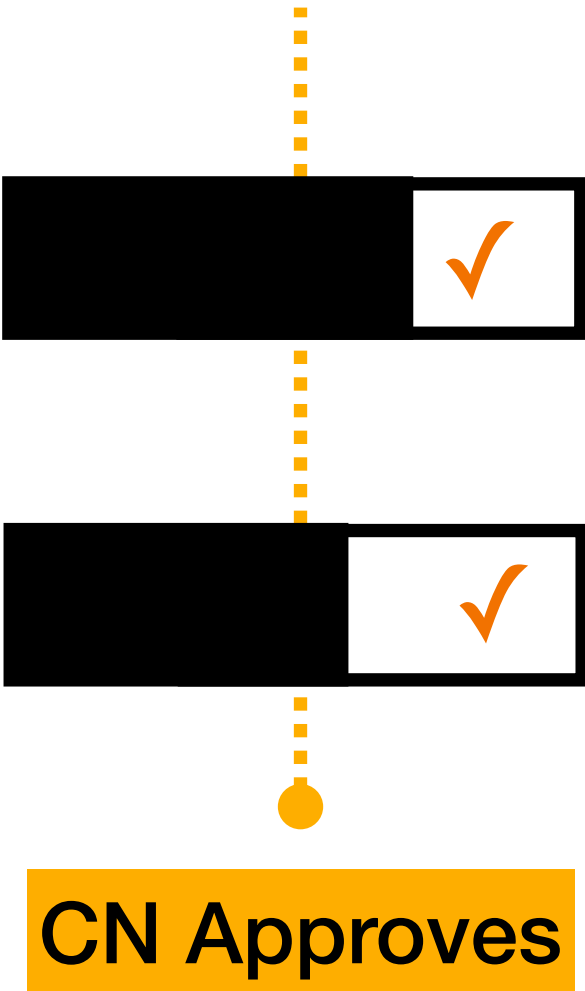
deep, persistent disagreement



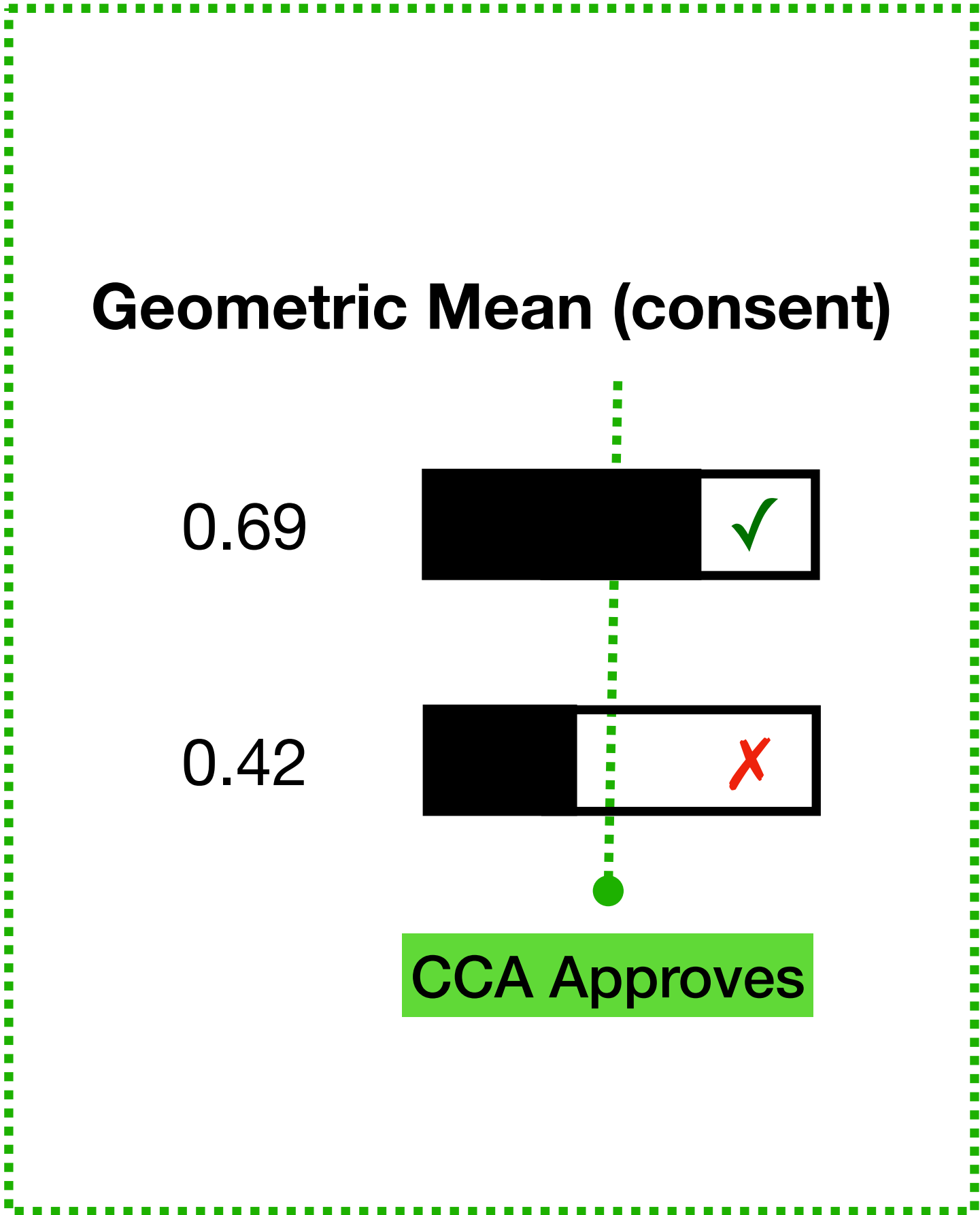
X's Community Notes sits at the left. CCA moves the decision right.

How the Math Moves

Note type	Group A	Group B	Average
both accept	70 %	68 %	69 %
one-sided	90 %	20 %	55 %



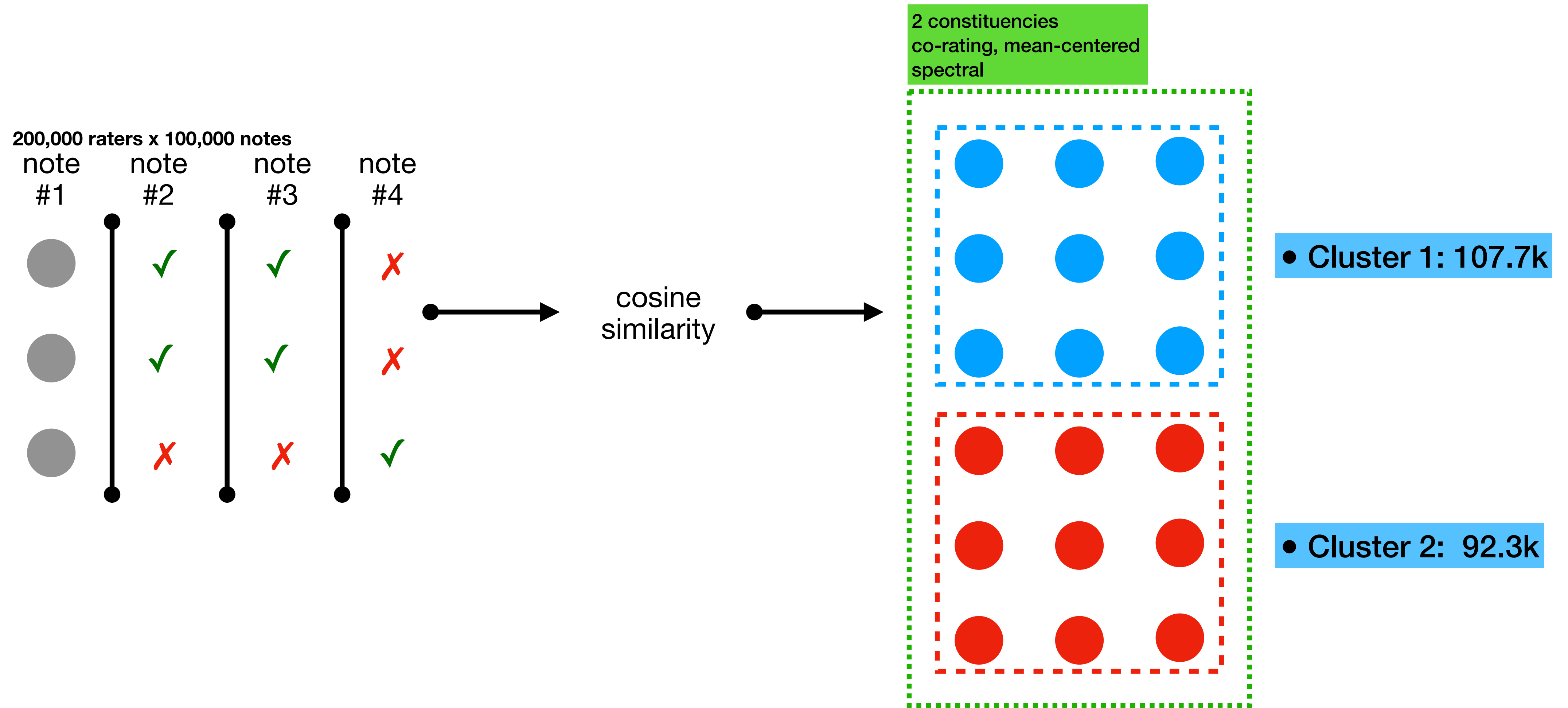
We call this a soft veto.



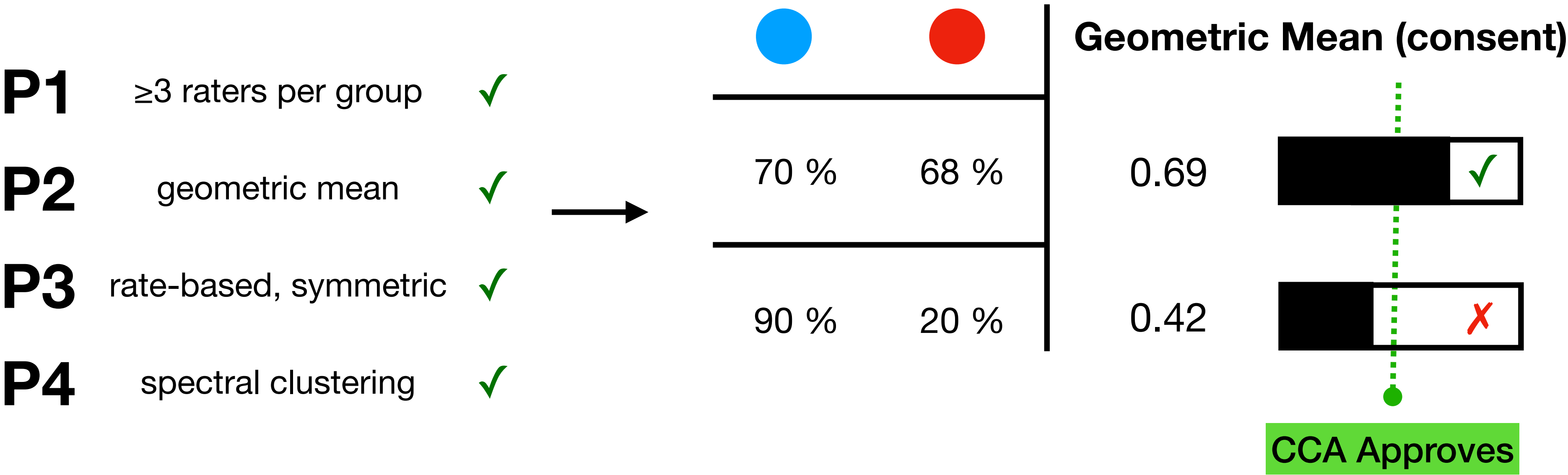
Our Approach

	principle	where it comes from	how we are doing?
P1	every group must be present	quorum / double-majority	≥ 3 raters per group
P2	one group's "no" can't be bought off	mutual veto / soft veto	geometric mean
P3	groups enter equally	parity rules	rate-based, symmetric
P4	groups come from behaviour	representative social choice	spectral clustering

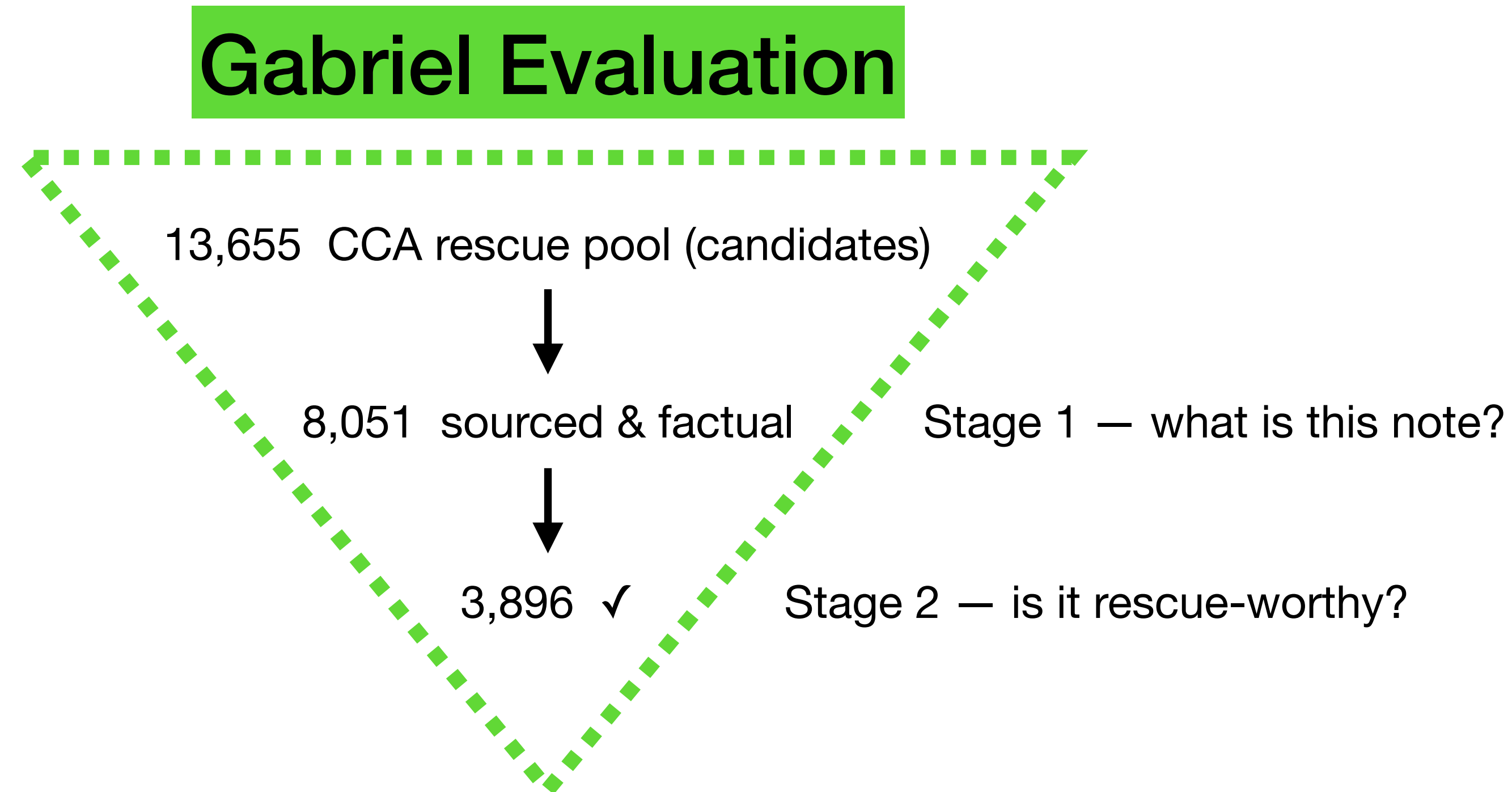
How We Are Clustering



How We Are Choose Notes



How We Are Evaluate Notes



This mirrors X's own emerging stack: a scoring layer finds candidates, and a content-understanding layer validates them.

The rescue is structural: CCA does the lifting. The LLM only checks that it is a real, sourced fact.

Thank You